

Visual Navigation Settings

Main controller settings includes:

- GPS Configuration
- Compass Calibration
- Altitude and Position Hold
- Waypoint Navigation
- Obstacle Avoidance
- Camera and Gimbal Settings

Let us learn about each in detail.



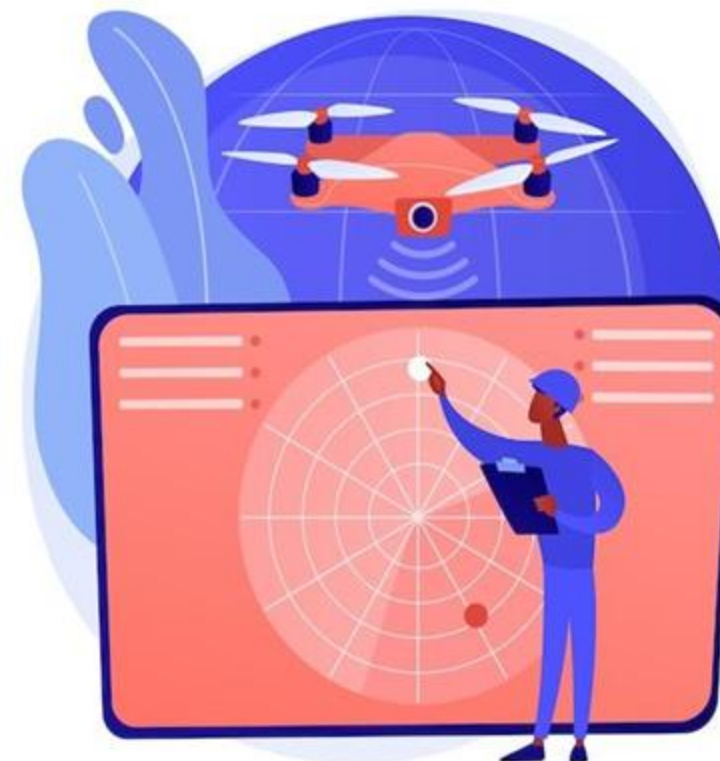
Alt text: Navigation setting

GPS Configuration

GPS (Global Positioning System) settings are crucial for accurate positioning and navigation.

Key considerations include:

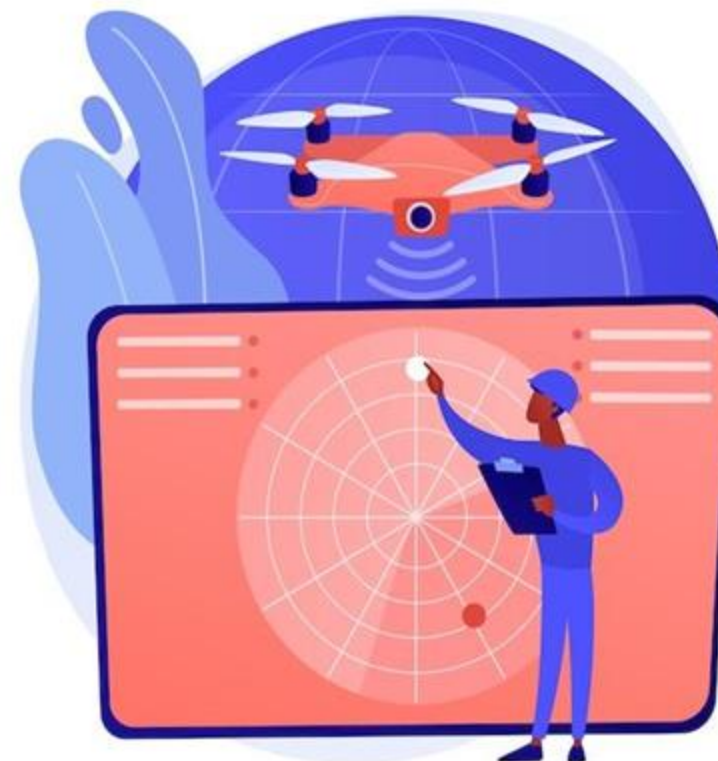
- **Accuracy Settings** Ensures the GPS system provides high-precision location data, often achieved through differential GPS (DGPS) or real-time kinematic (RTK) corrections.



Alt text: Military drone abstract concept illustration

GPS Configuration

- **Update Rate** This gives frequency at which GPS data is updated, impacting the responsiveness of navigation.
- **Failsafe Settings** These are predefined actions the drone should take if it loses GPS signal, such as returning to home (RTH) or hovering in place.



Alt text: Military drone abstract concept illustration

Compass Calibration



Alt text: Remote control flying a drone

The compass provides heading information, which is vital for accurate navigation.

Calibration involves:

- **Eliminating Magnetic Interference**
Ensuring there are no magnetic materials nearby that could distort the compass readings.

Compass Calibration



Alt text: Remote control flying a drone

- **Calibration Procedures** Often involve rotating the drone in different orientations to allow the compass to map the local magnetic field.
- **Regular Re-calibration** Needed when operating in different environments or after changes to the drone's hardware.

Altitude and Position Hold

Altitude and position hold involves:

- **Altitude Hold** Setting altitude hold parameters allows the drone to maintain a constant altitude, even in windy conditions or during manual control.
- **Position Hold** Configuring position hold settings enables the drone to maintain a fixed position, even if there are slight disturbances from wind or other factors.
- **Loiter Mode** Loiter mode allows the drone to hover in a fixed position, maintaining both altitude and position. This is useful for aerial photography or inspection tasks.



Alt text: Drone position

Waypoint Navigation



Alt text: Waypoint navigation

Waypoint navigation allows drones to fly autonomously along a predefined path.

Key elements include:

- **Waypoint Configuration** Setting GPS coordinates for each waypoint, including altitude, speed, and any specific actions at each point.

Waypoint Navigation



Alt text: Waypoint navigation

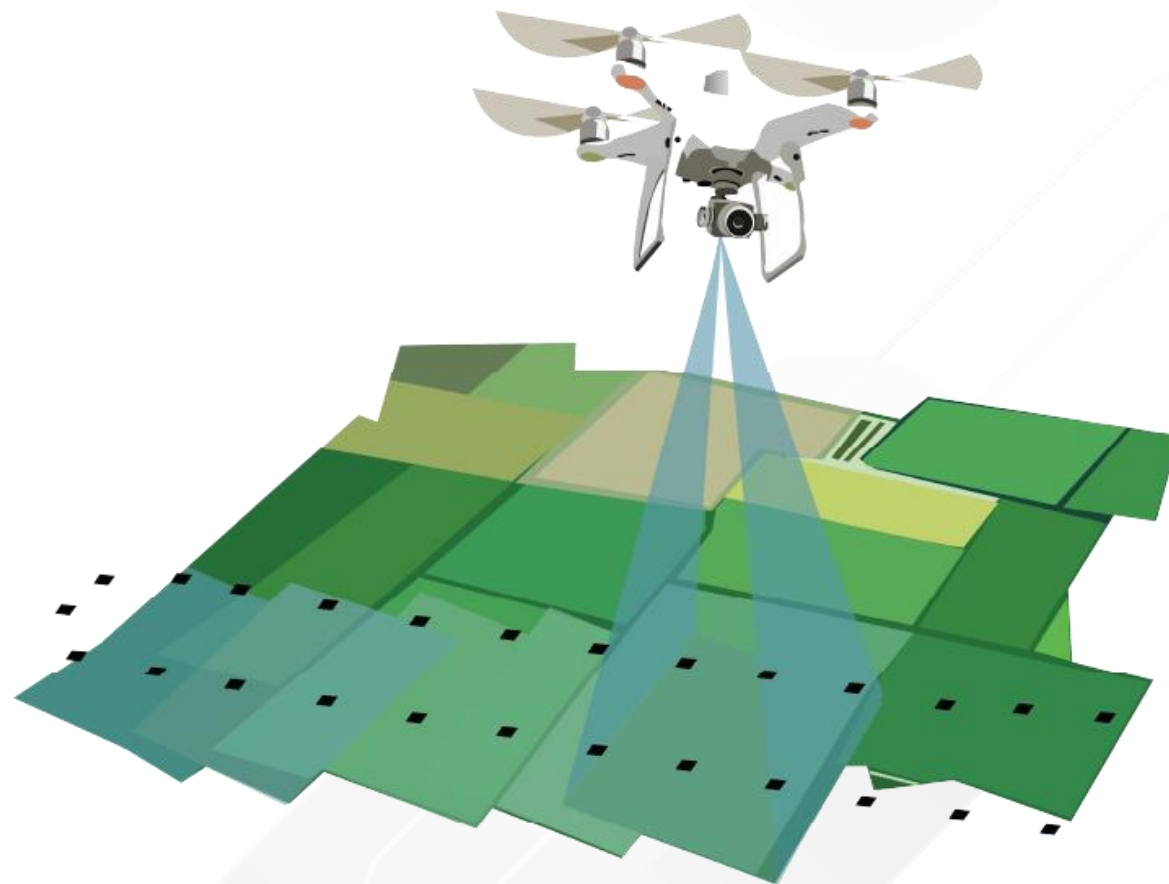
- **Path Smoothing** Algorithms to create smooth transitions between waypoints, avoiding sharp turns that could destabilise the drone.
- **Loiter and Hover Commands** Algorithms to create smooth transitions between waypoints, avoiding sharp turns that could destabilise the drone.

Obstacle Avoidance

Lidar or Ultrasonic Sensors:

Enabling and configuring these sensors allows the drone to detect obstacles in its path and take evasive action.

Lidar sensors provide more accurate and longer-range detection, while ultrasonic sensors are more suitable for short-range obstacle avoidance.



Alt text: Drone lidar

Camera and Gimbal Settings



Alt text: Drone camera

The optimal camera and gimbal settings for a drone depend on the specific task or application.

Key elements include:

- **Camera Parameters** Configuring camera settings such as resolution, frame rate, exposure, and white balance allows for optimal image quality based on the specific application.

Camera and Gimbal Settings



Alt text: Drone camera

- **Gimbal Stabilization** Enabling gimbal stabilisation ensures that the camera remains steady, even during turbulent flight conditions.
- **Camera Control Options** Configuring camera control options allows for remote control of the camera, such as zooming, focusing, and taking photos or videos.